



Calculus 1M – Final A (Summer 2021)

INSTRUCTIONS: Write Your Answers to each problem on the **SAME PAGE** in the blank space provided. **NO Electronic Devices or Notes Allowed.**

There are two parts of the exam:

- **Part 1:** Write the complete solution like in the Written HW.
- **Part 2:** Fill the answer box like in the Webwork HW.

Your Name: _____

Student ID: _____

Good luck! 😊

Part 1: PROBLEMS 1-6, Show All Work For a Full Credit.

1. (10 points) Find

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x+2} \right)^x.$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \ln \left(\frac{x}{x+2} \right)^x &= \lim_{x \rightarrow \infty} x \ln \left(\frac{x}{x+2} \right) = \lim_{x \rightarrow \infty} \frac{\ln \left(\frac{x}{x+2} \right)}{\frac{1}{x}} \\ &\stackrel{L^1H}{=} \lim_{x \rightarrow \infty} \frac{\frac{x+2}{x} \cdot \frac{2}{(x+2)^2}}{-\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} - \frac{2x}{x+2} = -2 \end{aligned}$$

Thus,

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\frac{x}{x+2} \right)^x &= \lim_{x \rightarrow \infty} e^{\ln \left(\frac{x}{x+2} \right)^x} \\ &= e^{\lim_{x \rightarrow \infty} \ln \left(\frac{x}{x+2} \right)^x} = e^{-2}. \end{aligned}$$

2. (10 points) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} ax^2 + b & \text{if } x \leq 1 \\ \frac{4}{\sqrt{x}} + 1 & \text{if } x > 1 \end{cases}$$

Find the values of the real numbers a and b such that the function is differentiable in its domain.

Hint: A relation between differentiability and continuity at the point could be used to find the second parameter.

f is diff. on $(-\infty, 1) \cup (1, +\infty)$, so we just need to look at differentiability at $x=1$.

If f is diff. at 1 then f is continuous at 1, so

$$f(1) = a + b = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 5$$

so f is continuous at 1 $\Leftrightarrow a + b = 5$.

$$\begin{aligned} \lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1} &= \lim_{x \rightarrow 1^-} \frac{ax^2 + 5 - a - 5}{x - 1} \\ &= \lim_{x \rightarrow 1^-} \frac{a(x^2 - 1)}{x - 1} = \lim_{x \rightarrow 1^-} a(x + 1) \\ &= 2a. \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1} &= \lim_{x \rightarrow 1^+} \frac{\frac{4}{\sqrt{x}} + 1 - 5}{x - 1} = \lim_{x \rightarrow 1^+} \frac{4 - 4\sqrt{x}}{\sqrt{x}(x - 1)} \\ &= \lim_{x \rightarrow 1^+} \frac{4(1 - \sqrt{x})}{\sqrt{x}(\sqrt{x} - 1)(\sqrt{x} + 1)} \\ &= \lim_{x \rightarrow 1^+} \frac{-4}{\sqrt{x}(\sqrt{x} + 1)} = -2. \end{aligned}$$

so f is diff. at $x=1 \Leftrightarrow a = -1$ and $b = 6$.

3. (10 points) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function.

- Find the *Absolute Minimum* and *Absolute Maximum* of the function on $[-2, 1]$.
- Find all the *Local Extrema* of this function on $\mathbb{R}$.

For Calculus 1M ONLY:

$$f(x) = x^3 - \frac{3}{2}x^2 - 6|x| + 10$$

f IS DIFF. ON $\mathbb{R} \setminus \{0\}$ WITH

$$f'(x) = \begin{cases} 3x^2 - 3x - 6 & \text{if } x > 0 \\ 3x^2 - 3x + 6 & \text{if } x < 0 \end{cases}$$

$$\text{AND } f'(x) = 0 \Leftrightarrow \begin{cases} x^2 - x - 2 = 0 & \text{AND } x > 0 \\ x^2 - x + 2 = 0 & \text{AND } x < 0 \end{cases} \Leftrightarrow x = 2.$$

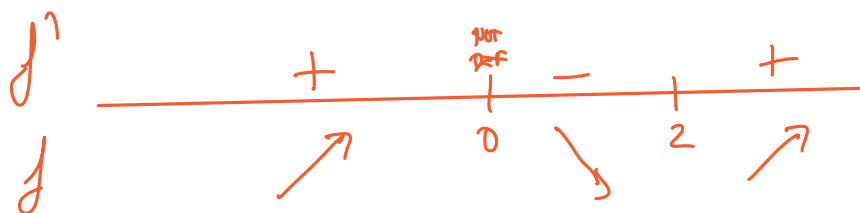
(a) SO THE CRITICAL POINTS OF f ON $[-2, 1]$ ARE $-2, 1$ (ENDPOINTS) AND 0 .

$$f(-2) = -8 - 6 - 12 + 10 = -16 \quad \leftarrow \text{ABSOLUTE MIN}$$

$$f(1) = 1 - \frac{3}{2} - 6 + 10 = \frac{7}{2}$$

$$f(0) = 10 \quad \leftarrow \text{ABSOLUTE MAX}$$

(b) THE POSSIBLE LOCAL EXTREMA OF f ON $\mathbb{R}$ ARE AT THE CRITICAL POINTS: $x = 0, 2$. WE LOOK AT THE SIGN OF f' TO CLASSIFY THEM:



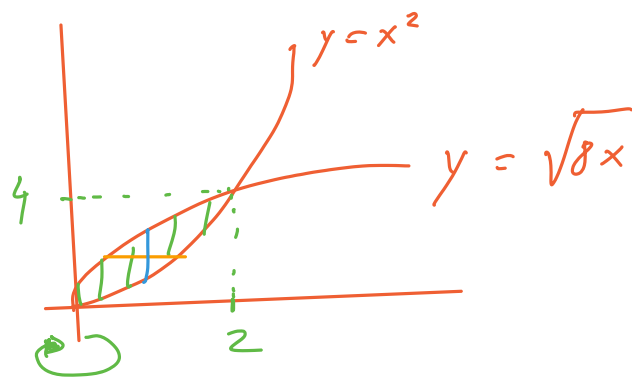
NOTE THAT f IS CONTINUOUS SO BY THE 1ST DERIVATIVE TEST,
 $x = 0$ IS A LOCAL MAX AND
 $x = 2$ IS A LOCAL MIN.

4. (10 points) Find the volume of the solid generated by rotating the region bounded by the given curves around the y -axis.

For Calculus 1M ONLY: $y = x^2, y = \sqrt{8x}$.

$$x^2 = \sqrt{8x} \Leftrightarrow x^4 = 8x \quad \text{AND } x \geq 0$$

$$\Leftrightarrow x = 0 \quad \text{OR } x = 2$$



$$V = \pi \int_0^4 \left((\sqrt{y})^2 - \left(\frac{y^2}{8} \right)^2 \right) dy = \pi \int_0^4 y - \frac{1}{64} y^4 dy$$

$$= \pi \left[\frac{y^2}{2} - \frac{1}{64} \frac{y^5}{5} \right]_0^4 = \pi \left(\frac{16}{2} - \frac{4^5}{64 \times 5} \right)$$

$$= \pi \left(\frac{80}{10} - \frac{32}{10} \right) = \frac{24}{5} \pi$$

OR

$$V = 2\pi \int_0^2 x (\sqrt{8x} - x^2) dx = 2\pi \int_0^2 \sqrt{8} x^{\frac{3}{2}} - x^3 dx$$

$$= 2\pi \left[\sqrt{8} \frac{2}{5} x^{\frac{5}{2}} - \frac{1}{4} x^4 \right]_0^2 =$$

$$= 2\pi \left(\frac{32}{5} - 4 \right) = \frac{24}{5} \pi$$

5. (10 points) Test the given problems for the convergence or divergence. Specify the test used. If it converges, find the value.

For Calculus 1M ONLY:

a) $\int_1^2 \frac{1}{x \ln x} dx$.

b) $\sum_{n=0}^{\infty} \frac{(-2)^n}{5^{n+1}}$.

(a) $\frac{1}{x \ln x}$ IS NOT DEFINED FOR $x=1$. LET $1 < \epsilon \leq 2$.
 THEN $\int_{\epsilon}^2 \frac{1}{x \ln x} dx = \int_{\ln \epsilon}^{\ln 2} \frac{1}{u} du = \left[\ln u \right]_{\ln \epsilon}^{\ln 2}$
 $\downarrow$
 $u = \ln x$
 $du = \frac{1}{x} dx$
 $= \ln(\ln 2) - \ln(\ln \epsilon)$.

$\lim_{\epsilon \rightarrow 1^+} \int_{\epsilon}^2 \frac{1}{x \ln x} dx = \lim_{\epsilon \rightarrow 1^+} \ln(\ln 2) - \ln(\ln \epsilon) = +\infty$.

SO THE INTEGRAL DIVERGES

(b) $\sum_{n=0}^{\infty} \frac{(-2)^n}{5^{n+1}} = \frac{1}{5} \sum_{n=0}^{\infty} \left(-\frac{2}{5}\right)^n = \frac{1}{5} \frac{1}{1 - (-\frac{2}{5})}$
 $= \frac{1}{5} \frac{1}{1 + \frac{2}{5}} = \frac{1}{5+2} = \frac{1}{7}$.

6. (10 points)

- a) Write the Taylor polynomial of degree 3 for the function $f(x) = 32\sqrt{x}$ around $a = 4$.
- b) Use the second degree polynomial (from the part above) to approximate $\sqrt{2}$.
- c) Estimate the error of an approximation.

NOT COVERED THIS YEAR.

Part 2: PROBLEMS 7-12 Fill the Answer Box. Explanations are NOT required and NOT graded.

7. (8 points) Find the following limits:

$\lim_{x \rightarrow 0^+} \frac{ \sin x }{x} = \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$	$\lim_{x \rightarrow 0^-} \frac{ \sin x }{x} = \lim_{x \rightarrow 0^-} \frac{-\sin x}{x} = -1$
$\lim_{t \rightarrow 0} \frac{\tan 3t}{5t} = \frac{3}{5}$	$\lim_{x \rightarrow -\infty} \frac{3 + \sqrt[3]{x} - 5x}{-4 + 2x + x^{2/3}} = -\frac{5}{2}$

8. (10 points) Find $\frac{dy}{dx}$ using explicit or implicit differentiation.

If $y = \left(\frac{e^x - x}{1 + \sin x} \right)^3$ then $\frac{dy}{dx} = 3 \left(\frac{e^x - x}{1 + \sin x} \right)^2 \frac{(e^x - 1)(1 + \sin x) - (e^x - x) \cos x}{(1 + \sin x)^2}$
If $xy^2 = \ln x$ then $\frac{dy}{dx} = \frac{1}{2x^2y} - \frac{y}{2x}$

9. (5 points) Let

$$f(x) = F'(x) = \cos 2x - 2x \sin 2x$$

$$F(x) = \int_1^x f(t) dt = x \cos 2x.$$

Find $f\left(\frac{\pi}{2}\right) = (x \cos 2x)' \Big|_{x=\pi/2} = \cos \pi - 2 \cdot \frac{\pi}{2} \sin \pi = -1$

10. (9 points) Let a rational function is $f(x) = \frac{4x^2 - 9x + 3}{x(x-1)^2}$.

Write the partial fraction decomposition of $f(x)$.

$$f(x) = \frac{3}{x} + \frac{1}{x-1} - \frac{2}{(x-1)^2}$$

Evaluate $\int f(x) dx = 3 \ln |x| + \ln |x-1| + 2 \frac{1}{x-1} + C$

11. (4 points) Suppose that $\sum_{n=0}^{\infty} c_n x^n$ converges when $x = -3$ and diverges when $x = 6$. Answer "Converges" or "Diverges" for the following series.

$\sum_{n=0}^{\infty} c_n$ CONVERGES	$\sum_{n=0}^{\infty} c_n 9^n$ DIVERGES
$\sum_{n=0}^{\infty} c_n (-8)^n$ DIVERGES	$\sum_{n=0}^{\infty} (-1)^n c_n 2^n$ CONVERGES

12. (4 points) Let the series is

$$\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{2^n (n+3)}$$

Find the Radius of Convergence $R = 2$

Find the Interval of Convergence $(-2, 2]$